

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination (EC/ME/CL/CE/IT/EE)

May 2012

EE-102 Fundamentals of Electrical Engineering

Date: 15.05.2012, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

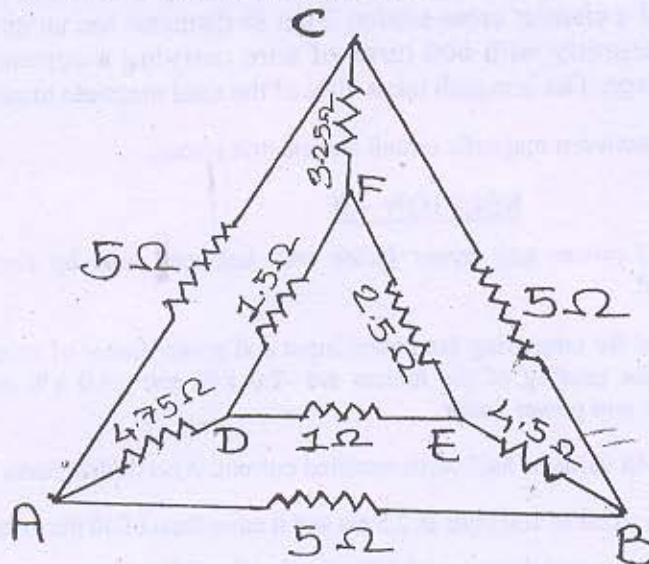
Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Find the resistance between the points A and B.

[05]



(b) State Ohm's law of electric circuit. Also give its limitations.

[02]

Q - 2 (a) An uncharged capacitor C Farads in series with a resistor of R ohms is connected to a constant supply voltage V volts. Derive an expression for the voltage across the capacitor in terms of time. Sketch the variation of voltage against time.

[07]

(b) A capacitor is charged to a d.c. source through a resistor of $1\text{ M}\Omega$. If in 1 second, the potential difference across the capacitor reaches 80% of the final value. Calculate the capacitance of the capacitor.

[03]

(c) State and explain Kirchhoff's Laws.

[04]

OR

Q - 2 (a) Derive an expression for the capacitance of a capacitor when it contains partly air and partly dielectric of ϵ_r .

[04]

(b) Two perfectly insulated capacitors are connected in series. One is an air capacitor with a plate area of 100 cm^2 , the plate being 1 mm apart; the other has a plate area of 10 cm^2 , the plates being separated by a solid dielectric 0.1 mm apart with a relative permittivity of 5. Find the voltage across the combination if the potential gradient in the air capacitor is 20 kv/cm .

[06]

- (c) State and explain Faraday's laws of electromagnetic induction. [04]
- Q - 3 (a) What are the factors on which the inductance of a coil depend? Derive the necessary expression for calculating the inductance of a coil. [07]
- (b) The force between two long parallel conductors is 15 kg/m. The conductor spacing is 10 cm. If one conductor carries twice the current of the other, calculate the current of each conductor. [05]
- (c) Define the following: [02]
1. Leakage coefficient. 2. Lenz's law. 3. Reluctance. 4. Permeance.

OR

- Q - 3 (a) Explain Hysteresis loss and Eddy current loss with necessary diagram. [06]
- (b) Determine M.M.F., Magnetic flux, Reluctance and Flux density in case of a steel ring 30 cm mean diameter and a circular cross-section 2 cm in diameter has an air gap 1 mm long. It is wound uniformly with 600 turns of wire carrying a current of 2.5 A. Neglect magnetic leakage. The iron path takes 40% of the total magnetomotive force. [06]
- (c) Compare dissimilarities between magnetic circuit and electric circuit. [02]

SECTION – II

- Q - 4 (a) Explain measurement of power and power factor with balanced load by two wattmeter method for inductive load. [05]
- (b) Two watt meters are used for measuring the power input and power factor of an over excited synchronous motor. If the reading of the meters are -2.0 kW and +7.0 kW respectively, Calculate the input power and power factor. [02]
- Q - 5 (a) Derive expression for RMS value of half-wave rectified current. Also define Form factor. [07]
- (b) A sinusoidal voltage has a value of 100 volts at 2.5 ms and it takes time of 20 ms to complete one cycle. Find the maximum value and time to reach it for the first time after zero. [07]

OR

- Q - 5 (a) Derive an expression for the instantaneous value of alternating emf in terms of its maximum value, angular frequency and time. [07]
- (b) A capacitor and a resistor are connected in series to an a.c. supply of 50 V, 50 Hz. The current is 2 A and the power dissipated in the circuit is 80 W. Calculate the values of R and C. [07]
- Q - 6 (a) Prove that the current in a purely capacitive circuit leads its voltage by 90° . [07]
- (b) Define Q-factor and also derive expression for Q-factor. [07]

OR

- Q - 6 (a) Compare series and parallel resonance. [07]
- (b) A series RLC circuit with $R=10\ \Omega$, $L=10\text{ mH}$ and $C=1\ \mu\text{F}$ has an applied voltage of 200 V at resonant frequency. Calculate the resonant frequency, the current in the circuit and voltage across the elements at resonance. Find also Q-factor and band width. [07]